03 · CONDITIONAL PATCHGAN ADVERSARIAL PIPELINE

Overcoming the Over-Smoothing Trap via Multi-Modal Discriminator Critique

Project: SAR-Guided Satellite Cloud Removal · ECRformer + Conditional PatchGAN
Authors: Agnal K V · Bivin Balu · Vaishnav T Prakash | **Supervisor:** Dr. Sabu M K
Department of Computer Applications, Cochin University of Science and Technology (CUSAT)

Document Purpose & Adversarial Objective

Standard deep learning architectures trained solely with pixel-level distortion losses (L_1, L_2, SSIM) inevitably suffer from **regression to the conditional mean**—producing blurry, lifeless surfaces under thick clouds. This document details how our **11-channel Conditional PatchGAN Discriminator** (D_ϕ) and **Opaque Cloud Cutout** work synergistically to restore realistic high-frequency ground detail (+142% sharpness gain), and justifies **why each adversarial design choice outperforms traditional GAN frameworks**.

1 The Core Problem: The "Over-Smoothing Trap"

When an optical satellite patch is covered by an opaque cloud, the true underlying surface could be a dark asphalt highway, a green lawn, or a terracotta building roof. Because regression losses ($L_1 = |P - Y|$) penalize large deviations heavily, the mathematically optimal strategy for the neural network is to predict the **average of all possibilities** (a flat greyish-brown smudge).

$$P^* = \arg\min_P \mathbb{E}[\|P - Y\|_1] = \text{Median}(Y|X) \implies \text{Severe Blurring} \tag{1}$$

Our metric measurements confirm this phenomenon: the baseline ECRformer model achieves high PSNR (30.20 dB), but its **Detail Ratio** (ratio of high-frequency gradient energy to real ground truth) plunges to **0.354** under dense overcast. To solve this, we introduce an adversarial game.

2 Architecture of the 11-Channel PatchGAN Critic

Instead of evaluating an image globally as "real" or "fake", our discriminator operates on local 16×16 spatial patches.

Discriminator Specifications (D_ϕ):

- **Input Dimensionality (11 Channels):**

$$X_D = \Big[S_1\,(\text{2ch}),\ S_{2,\text{cloudy}}\,(\text{4ch}),\ M\,(\text{1ch}),\ \text{Candidate}\,(\text{4ch})\Big] \in \mathbb{R}^{B\times 11\times 128\times 128} \tag{2}$$

  where Candidate is either real ground truth Y or synthesized prediction P_{dir}.

- **Local Receptive Field (16×16 Patch Grid):** Rather than outputting a single scalar, D_ϕ outputs an 8×8 grid of local validity logits. Each logit evaluates whether a specific 16×16 ground patch matches natural optical texture and obeys the underlying SAR radar dielectric profile.

- **High-Pass Spectral Focus:** By restricting the receptive field to 16×16, the discriminator focuses exclusively on fine local textures (crop furrows, road lanes, building edges), leaving global color calibration to the L_1 and SDFL losses.

3 Training Objectives & Loss Dynamics

The adversarial framework alternates between training the Critic D and Generator G across two computational graphs.

3.1 1. Generator Adversarial & Feature Matching Loss

$$\mathcal{L}_{\text{adv}}(G) = \mathbb{E}_X \left[(D(X, P_{\text{dir}}) - 1)^2 \right] \quad \text{(Least Squares GAN penalty)} \tag{3}$$

$$\mathcal{L}_{\text{FM}}(G, D) = \sum_{l=1}^{L} \frac{1}{N_l} \left\| D^{(l)}(X, Y) - D^{(l)}(X, P_{\text{dir}}) \right\|_1 \quad \text{(Perceptual feature matching)} \tag{4}$$

3.2 2. The Composite Generator Loss Formulation

The complete loss driving ECRformer's backpropagation is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{L1} + \lambda_{\text{SDFL}} \mathcal{L}_{\text{SDFL}} + w_{\text{adv}} \mathcal{L}_{\text{adv}} + w_{\text{FM}} \mathcal{L}_{\text{FM}} + \lambda_{\text{corr}} \mathcal{L}_{\text{corr}} \tag{5}$$

where $w_{\text{adv}} = 0.04$ and $w_{\text{FM}} = 0.35$ maintain stable Nash equilibrium without gradient explosion.

4 Opaque Cloud Cutout (65% Modality Dropout)

In the SEN12MS-CR dataset, many training patches contain semi-transparent haze where road contours faintly leak through optical bands (correlation > 0.80).

The Modality Cheating Trap

If the network is allowed to see these faint leaked optical edges during training, it learns a "shortcut"—it simply amplifies the faint optical lines and completely ignores the Sentinel-1 SAR radar stream. Consequently, when presented with real-world thick, opaque clouds, the model completely fails.

How We Prevent Cheating: In 65% of training iterations (`prob=0.65`), we apply **Opaque Cloud Cutout**:

1. Identify cloudy pixels using $M > 0.5$.
2. Erase all optical ground textures under the mask using two consecutive passes of 15×15 average pooling.
3. Add realistic opaque cloud brightness, completely destroying optical edge information while maintaining valid $[0, 1]$ radiometric bounds.

This **forces** ECRformer to extract 100% of its structural boundaries from Sentinel-1 SAR radar!

5 Why We Used These Methods Over Others

1. Why Conditional PatchGAN over Standard Global GAN (e.g. DCGAN, WGAN)?

Alternative: Standard global discriminator that condenses the entire 128×128 image into a single scalar real/fake score.

- **Failure of Alternative:** Global discriminators suffer from mode collapse. When forced to critique an entire satellite scene at once, they prioritize large-scale visual harmony, often hallucinating entire fictional city blocks or ignoring subtle blurry roads.
- **Why Our Choice Wins:** PatchGAN operates as a local Markovian random field. It evaluates 16×16 patches independently. This forces the generator to produce crisp, high-frequency textural fidelity at the micro-scale, while the L_1 and SDFL terms ensure macro-scale geographic accuracy.

2. Why an 11-Channel Multi-Modal Discriminator over an Optical-Only Discriminator?

Alternative: Feeding only the 4-channel generated optical image P_{dir} to the discriminator.

- **Failure of Alternative:** An optical-only critic only asks: *"Does this patch look like plausible Earth terrain?"* The generator can easily trick it by generating a plausible forest over an area that is actually a seaport.
- **Why Our Choice Wins:** By concatenating S_1 SAR (VV, VH) directly into the discriminator input (11 channels), the critic explicitly cross-references the optical prediction against the true radar backscatter. If the generator hallucinates trees where the radar shows smooth water, the 11-channel critic penalizes it immediately.

3. Why Feature Matching Loss ($\mathcal{L}_{\text{FM}}$) over Pure Adversarial Loss Alone?

Alternative: Training with only the binary cross-entropy or least-squares adversarial loss.

- **Failure of Alternative:** Pure adversarial gradients are notorious for oscillating and causing high-frequency GAN "checkerboard" speckle noise.
- **Why Our Choice Wins:** Feature Matching ($\mathcal{L}_{\text{FM}}$) matches the multi-scale activation representations across intermediate layers of the discriminator ($D^{(l)}$). It acts as a natural, self-supervised perceptual loss that stabilizes generator gradients and completely suppresses adversarial artifacts.

4. Why the Sweet-Spot Model (Score: 0.5439) over Late-Epoch GAN Checkpoints?

Alternative: Running GAN training to 50 epochs and picking the final checkpoint.

- **Failure of Alternative:** Beyond epoch 25, the discriminator becomes overly aggressive, forcing the generator to inject unnatural high-contrast speckle grain into clear terrain.
- **Why Our Choice Wins:** We implemented a composite selection score:

$$\text{Score} = \text{EdgeCorr}_{\text{heavy}} + 0.5 \cdot \text{DetailRatio}_{\text{heavy}} - 5.0 \cdot \text{MAE}_{\text{all}} \tag{6}$$

The designated **Sweet-Spot Model** (Epoch 18, Score 0.5439) captures peak edge correlation (0.45+) and natural detail ratio (0.91) without any synthetic noise artifacts.

6 15-Second Viva Examiner Defense Quick-Sheet

Q: What is the Detail Ratio metric and why is it superior to PSNR for cloud removal?
Answer: Detail Ratio measures the ratio of high-frequency gradient energy between predicted and ground-truth optical images. PSNR rewards blurry conditional mean predictions; Detail Ratio specifically quantifies whether roads, coastlines, and field boundaries are sharp (1.00 = 100% natural crispness).

Q: Why is your PatchGAN discriminator conditional on Sentinel-1 SAR?
Answer: Conditioning on SAR radar ensures that the generated optical features are physically grounded in actual microwave surface roughness rather than hallucinated by the generator.

Q: What role does the Discriminator Warmup step play?
Answer: In the first 100 steps, only the Discriminator is updated. This establishes an accurate local texture critique before the Generator updates its weights, preventing early training instability.